Towards Automated Visual Inspection of Electrical Grid Assets for the Smart Grid - An Application to HV Insulators

Fernando Lopes[1,2]
[1]*Polytechnic University of Coimbra*
Rua da Misericórdia, Lagar dos Cortiços,
S. Martinho do Bispo, 3045-093 Coimbra, Portugal
flopes@isec.pt

Pedro Rocha[1,2]
[2]*Instituto de Telecomunicações - Coimbra*
Pinhal de Marrocos, Pólo II DEEC
3030-290 Coimbra, Portugal
pedro.rocha@co.it.pt

André Coelho[3]
[3]*EDP Labelec*
Inspeção de Ativos
2685-038 Sacavém, Portugal
andre.coelho@edp.com

***Abstract*—The globally growing demand for electric power, projected to double by 2050, requires extensive upgrades in Transport and Distribution (T&D) systems. Despite the need for new infrastructures to address this demand, the reliability of ageing electrical T&D systems remains a more critical concern, where power line insulators account for over 50% of T&D maintenance costs. To boost inspection efficiency, electric utilities are integrating remote inspection technologies into their operations. However, the processing of the growing volume of collected data is currently strongly limited by human interpretation tasks.**

This research evaluates the performance of computer vision based on deep learning models for automated visual inspection of electrical grid assets, validated with real-world data. The developed work presents an example application of automated visual inspection of HV insulators, detecting defects in disc insulators in visible light images, using two state-of-the-art deep learning models. YOLOv8s achieved a mAP@50 of 87.9%, while Faster R-CNN X101-FPN achieved 87.2% for the same metric. The findings highlight the advantages and limitations of automated visual inspection, enabling utility companies to benefit from higher efficiency inspection processes, reducing the costs and improving the reliability of electrical grid maintenance.

Assessing the performance and complexity of data-driven automated visual inspection techniques is crucial for developing streamlined models that effectively handle high data volumes, and that can evolve to real-time operation and integration in the monitoring and control functions of the smart grid or as a dynamic component of a digital twin of the grid.

***Index Terms*—Asset Inspection, HV Insulator, Computer Vision , Deep Learning, Smart Grid**

I. Introduction

There is a globally growing demand for electric power, projected to double by 2050, and exceeding 50,000 TWh annually. Transmission and Distribution (T&D) grid expansion must follow this growth, reaching over 160 million km [1]. In this huge infrastructure, only power line insulators account for over 50% of the maintenance costs [2]. In addition to this expansion, the need for the integration of renewable and distributed energy resources, to allow for the transition away from carbon-based fuels, requires further extensive upgrades in T&D systems [3] .

To address these needs, new infrastructures are required. Meanwhile, the reliability of the existing ageing electrical generation and distribution systems must be maintained - in the European Union, more than 50% of the grid has been in operation for over 20 years, what is approximately half of its average lifespan. In addition, political instability drove the European Commission REPowerEU plan to implement new energy policies at an accelerated pace, putting further pressure on T&D infrastructures, that need to guarantee a reliable dependency on a stable and uninterrupted supply of electricity. The EU DSO Entity – the association representing the DSOs at European level has been contributing to the EU energy policy implementation and has analysed emerging tendencies through the Distribution System Observatory [4].

To ensure that modern society can continue to depend on a stable and uninterrupted supply of electricity, regular maintenance, visual inspections, and predictive maintenance techniques are being employed to identify and address potential issues before they lead to failures. Utility companies employ a range of strategies to maintain the availability and reliability of electric power distribution systems [5]. In recent decades, real-time remote monitoring of critical equipment has been implemented, enabling operators to identify anomalies without requiring physical presence. Combining sensor information and advanced data analysis, the aim is to predict equipment failures from performance trends and operating conditions, enabling timely interventions, reducing downtime and cost [6].

Insulators account for only 5–8% of installation costs, yet they incur over 50% of maintenance expenses [2]. Collecting aerial in-person inspection images costs range from 150 to 800 €/km, while automated aerial inspections cost between 40 and 150 €/km [7]. Electric utilities are collecting visual data as part of their inspection process of electrical infrastructures, to detect potential flaws and to make informed decisions about maintenance prioritization. However, the increasing volume of collected data, especially images, during infrastructure inspections, together with its processing, has a bottle-neck in the human interpretation tasks, mainly because escalating this pipe-line element trough training is timely expensive.

Advancing these technologies to a comprehensive and reliable level for widespread adoption and practical implementation through commercial applications requires further research and development efforts.

The main objective of this work is to contribute to the evaluation of the potential value of an automated system for visual inspections of electrical grid assets using a computer vision system based on state-of-the-art (SOTA) deep learning models, to be used as a tool for electrical infrastructure maintenance management. For this purpose, an analysis of two SOTA object detectors using high-resolution images of HV insulators, assessing their strengths, weaknesses, and performance differences, is conducted. High performing models can serve as a reference to develop low complexity models for edge processing and integrate the smart grid operation.

The remaining of this paper is organised as follows. In Section II we present a representative review of techniques proposed for automated visual inspection of assets, focusing on insulators. We consider both acedemic research and business innovation solutions. Section III characterises the investigated models and dataset organisation. Both architecture and training are described, for a one-stage and for a two-stage detectors. Quantitative detection evaluations are presented in Section IV while a qualitative evaluation and a discussion of results are carried out in Section V. Conclusions of this work and directions for future research are presented in Section VI.

II. Related Work

Automated visual inspection workflows of high-voltage electrical infrastructure hold the promise of transforming traditional methods, providing safety, quality, and efficiency benefits to both utilities and consumers. Developing automated inspection technologies to handle real-world complexities requires large datasets, which are currently unavailable. The Electric Power Research Institute (EPRI) [8] seeks to address this data gap by creating and sharing datasets and by promoting software development competitions on open-source platforms, thereby fostering scientific research and business innovation.

A. Academic Research

Van Nguyen *et al.* [9] identified some of the main challenges in DL vision-based UAV inspection and proposed a set of solutions. Being the first challenge the lack of training data, they collected a total of 28,674 images with a resolution of 6048x4032. By cropping the bounding boxes, they managed to create three additional datasets, resulting in a total of 94,477 256x256 images. Recognizing the time-consuming nature of labeling a large dataset, they used data augmentation to address class imbalance, increasing the dataset size 12 times. To address the challenge of detecting small power components and small faults, they proposed a multi-stage component detection and classification pipeline, allowing for a 'zoom-in' inspection. Their best results in terms of mAP reached 81.3%, outperforming the other two methods in 7 out of 10 classes and achieving an inference speed of 300 images per minute.

Haiyan Cheng *et al.* [10] proposed a method for self-shattering defect detection of glass insulators based on spatial features using classical computer vision methods. Other researchers [11]–[13] have achieved similar outcomes, using comparable techniques in this context. Despite achieving over 90% accuracy, these techniques lack generalization across different insulator materials and colors. They focus solely on identifying the absence of a skirt in the insulator string, overlooking other prevalent issues like surface contamination.

To address uncertainty in manually labeled insulator datasets Zhiyong Dai [14] introduced a novel object detection DL-based methodology, for bounding box regression and uncertainty estimation, entitled YOLOD which integrates a Gaussian prior into the YOLOX model. By predicting bounding box uncertainties and using them as weighting factors in the Non-Maximum Suppression (NMS) process, YOLOD improves the robustness of insulator defect predictions. The comprehensive experiments achieved an average precision of 73.9%

High-resolution aerial images captured by UAVs with intricate backgrounds and small target detection present a critical challenge. Qiaodi Wen [5] overcame these complexities with two deep learning methodologies, Exact R-CNN and CME-CNN, both based on Faster R-CNN. Additionally, CME-CNN introduces an insulator mask extraction network to eliminate background interference and employs Exact R-CNN for defect detection. Experimental results demonstrate the superior effectiveness of these methods, with CME-CNN-ResNet50 achieving an average precision of 88.7%.

B. Business Innovation

Companies like eSmart Systems Grid Vision [15], that focus on managing asset information and conducting virtual inspections for T&D, performed a comparative analysis of two modes of visual inspection for overhead transmission lines: Virtual inspections, using high-resolution images from unmanned aircraft systems; And foot patrols involving systematic on-site inspections, documented on field computers. Results indicate that virtual inspections detected over 60% more defects than foot patrols. Certain defect types are more easily detectable in virtual inspections due to their top-down views, capturing components that might be challenging to spot during foot patrols. Conversely, foot patrols with a bottom-up view, excel in identifying other defects.

Noteworthy Inspect [16] integrates CV cameras, edge computing devices and proprietary AI models onto existing fleet vehicles without any intrusive modifications, to detect and geolocate utility poles during routine operations, capturing high-resolution images for each identified pole. These images serve as inputs for proprietary machine learning models, enabling diverse functionalities such as defect identification, asset inventories, attachment distance assessment, lighting evaluation, vegetation analysis, as-built assessments, and more. By combining stereo vision, point clouds and embedded neural networks, their technology produces highly accurate 3D reconstruction images, providing precision comparable to LiDAR technology but without its typical time and cost requirements.

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III. Methodology

This work focuses on the core of an automated inspection pipeline, receiving high resolution RGB images, collected by UAVs in real life scenarios of electric power T&D, containing different types and amounts of HV insulators, and generating a localization bounding box with a respective class of the predicted defect. The assets addressed in this study are the ceramic discs of each *Cap and Pin* insulator. Apart from the default background class, the implemented model classifies each insulator as either functional (good), defective disc (broken), or defective shell (flashover).

A. Dataset

To conduct our experiments, we used an organized Insulator Defect Image Dataset (IDID), released by the EPRI [8] and also published on *IEEE DataPort* [17], that comprises 1600 high-resolution images, accommodating 17116 annotated instances (78% good, 15% flashover and 7% broken).

Our observations suggest that the 1600 annotated image files, contain 400 unique images along with 3 mirrored copies. This information led us to avoid performing a random validation dataset split as has been done for example in [18], and not explicit in other cases [19] [20]. Creating a validation directory, carefully selecting 15% of the annotated images, ensured that no mirrored copies of the training dataset were included, ensuring the validation results integrity.

Upon closer examination of some images and their annotations, such as the one presented in Figure 1, it is noteworthy that, disc insulators cropped by the edge of the image, and *good* insulator discs, adjacent to *broken* ones, are not annotated. This decision potentially arose from the overlapping context of the bounding boxes.



Fig. 1. Example of missing annotations in the annotated IDID

Non-annotated data was used for test inference with supervised evaluation, including the other 88 images available in the IDID, and a sample of 24 images of MV power lines, from a large private dataset collected by EDP LabElec [21].

B. Architecture and Training

One of the goals of this work is to compare SOTA models. To achieve this, we employed the YOLOv8 one-stage detector and the Faster R-CNN two-stage detector using the Ultralytics [22] and the Detectron2 [23] high-level frameworks, respectively, both built on top of PyTorch DL library.

The variants of YOLOv8 for object detection range from the *nano* to the *extra-large* model. For faster evaluation, the *small* version, named *YOLOv8s*, was used, without compromising the main conclusions. To better capture small defective visual features in the images, the input size of the model was adjusted, from the default configuration of 640 pixels, to 1024 pixels, and was not further increased due to computational constraints. The default configuration for data augmentation [24] was used, which randomly performs a variety of transformations, including adjustments to HSV (Hue, Saturation, Value), image translation, scaling, horizontal flipping, among others. The batch size was set to 32. The LR was scheduled to start at 0.01 and decrease gradually by a factor of 1000 by the end of the training, which was set to 100 epochs.

For the two-stage detector, the X101-FPN variant was employed. This Faster R-CNN [25] model combines a backbone of 101 layers of ResNetX [26] with standard convolutional and fully connected heads for mask and bounding box prediction, respectively. In this implementation, the default input size configuration was used, that resizes the shortest edge of the input image from a set o values, ranging between 640 and 800, ensuring that the resized longest edge does not exceeds 1333 pixels. The default configuration for data augmentation was also used, which randomly flips the input image horizontally with a probability of 0.5. Since it is a two-stage detector, the batch size configuration was set to 4 images per batch, with a batch of 64 region proposals generated per image. The learning rate (LR) was scheduled to start at 0.01 and decrease by a factor of 10 in each of the 5 stages of 1000 iterations.

IV. Results

This section presents quantitative evaluations from YOLOv8 and Faster R-CNN implementations using standard metrics, including mAP and mAR, across typical IoU thresholds. The presented results were automatically obtained by inferring the validation dataset split, as it contains annotations.

The information in Figure 2 reveals that the YOLO model has successfully converged along the training iterations. Since the model has been loaded with pre-trained weights, the bounding box regression shows only a slight improvement, and more augmentation might not be as effective as focusing on increasing data quantity and quality. The most pronounced improvement is in the class attribution, as this represents novel information. Despite a consistent decrease in training loss, validation loss did not improve after epoch 56, indicating overfitting. Training was consequently stopped at epoch 71, since the patience was set to 15 epochs without validation loss improvement and the 56th model weights were used.

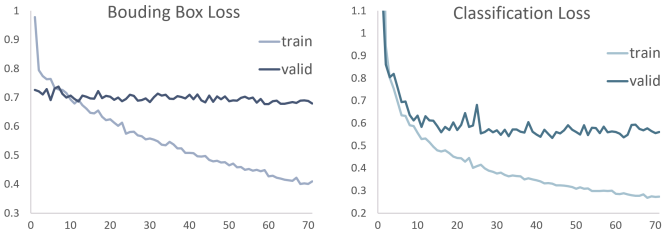


Fig. 2. Evolution of train and valid losses during YOLOv8s fitting

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Figure 3 presents the Precision and Recall as a function of the confidence threshold level, including the curves maximum points. The *good* class has the lowest precision but the highest recall. In contrast, the other classes have lower recall but high precision. For instance, since the *broken* class represents less than 10% of the instances but has highly distinctive features, this might indicate that the model is highly conservative in its predictions, which might not be a significant problem, in a scenario with multiple images of the same asset.

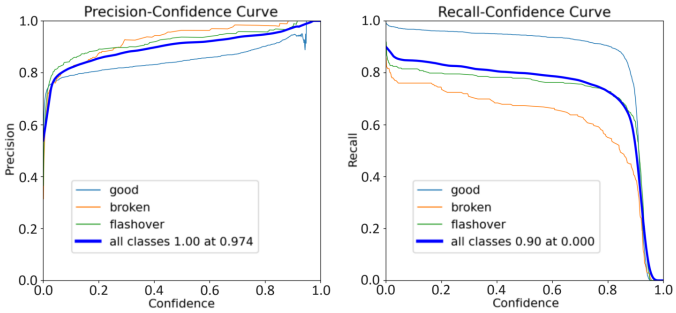


Fig. 3. Precision and Recall curves on the trained YOLOv8s

Observing the confusion matrix, in Figure 4, with absolute and normalized values, respectively, both computed at a default confidence threshold of 0.25, it is evident that several instances, particularly in the *good* class, have several predictions where the ground truth label is the background. One justification for this metric is that not all instances are annotated, as previously discussed and shown in Figure 1.

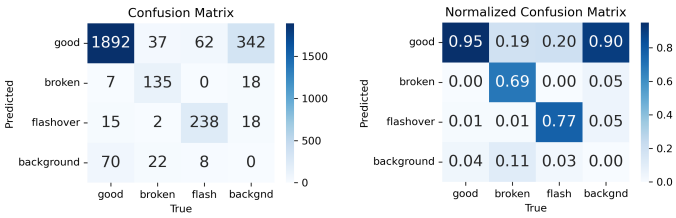


Fig. 4. Trained YOLOv8s confusion matrices

The standard performance metrics, easy accessible from Ultralytics APIs, are presented in Table I. The noticeable performance decay from the mAP50 to the mAP50-95 indicates that the bounding box alignment might be suboptimal. However, to rely heavily on this metric it is imperative that geometric precision of the annotations are considered trustworthy.

TABLE I
CLASS-WISE VALIDATION PERFORMANCE ON YOLOV8S

Class	Instances	Box P	Box R	mAP50	mAP50-95
all	2488	0.915	0.799	0.879	0.706
good	1984	0.844	0.945	0.929	0.789
broken	196	0.962	0.673	0.827	0.609
flashover	308	0.937	0.777	0.880	0.719

In the trained Faster R-CNN model, as shown in Table II, the precision is relatively higher than the recall. Class-wise analysis revealed high precision for the common class. The notable decrease in mAP@0.75 IoU in the *broken* class may be attributed to the variations in bounding box regression results caused by sharp edges. As mentioned earlier, the trustworthiness of these metrics is closely linked to the trustworthiness of the annotations, which requires a deeper analysis of the dataset. In this scenario of multiple instances of an object class in an image, setting the maximum detections to 1 severely limits the model's ability to recall all instances of objects in the image, predicting only the top score bounding box. By looking at recall in the initial detection, it is not surprising that the *broken* class has the higher value and the *good* class the lowest, since they are the lower and the higher representative class, respectively. In the defective classes, within the first 10 detections, the model identifies a significant portion of all the objects it will eventually detect. An exception to this trend is observed in the *good* class, which still experiences a significant increase in recall after the 10th detection. Primarily because most images contain more than 10 good insulators discs.

TABLE II
CLASS-WISE VALIDATION PERFORMANCE ON FASTER R-CNN

Metric	IoU	max Dets	Class			
			All	Good	Broken	Flash
mAP	@0.50	100	0.872	0.943	0.830	0.842
	@0.75	100	0.783	0.932	0.586	0.831
	@0.50:0.95	100	0.672	0.777	0.558	0.682
mAR	@0.50:0.95	100	0.761	0.847	0.664	0.772
	@0.50:0.95	10	0.714	0.723	0.664	0.756
	@0.50:0.95	1	0.283	0.099	0.483	0.266

A direct comparison reveals that YOLOv8 has surpassed Faster R-CNN in the most demanding metric (mAP50-95) across all classes. However, it's not always the case. A few examples exist, where one outperforms the other. To compare the validation results obtained in this work with those obtained in other studies that used the same dataset, a more common metric was also used, the mAP50, as other metrics were not available in the assessed literature. Table III summarizes the validation results of the Mean Average Precision at the 0.50 IoU threshold level in a class-wise comparison.

TABLE III
CLASS-WISE VALIDATION MAP@0.50 IOU COMPARISON

mAP	Model	Class			
		All	Good	Broken	Flashover
@0.50	YOLOv8s	**0.879**	0.929	0.827	**0.880**
	Faster R-CNN	0.872	**0.943**	**0.830**	0.842
	ML YOLOv5	0.776	0.834	0.748	0.745
	YOLOv5	0.327	0.760	0.000	0.220

This summary highlights the marginal superiority of the Faster R-CNN model in the *good* and *broken* classes, indicating that, while YOLO may achieve better bounding box regression accuracy, Faster may exhibit higher classification accuracy in some cases. Additionally, the results of the models implemented in this study outperform others found in the literature, more precisely, the results obtained by Das *et al.* [19], using YOLOv5, and the results obtained using a new variant named ML YOLOv5, proposed by Wang *et al.* [20].

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V. Evaluation and Discussion

By observing the inferred images in the testing dataset, which share similar characteristics to the annotated dataset, it is clear that both of the models (YOLOv8s in Figure 5 and Faster R-CNN in Figure 6), successfully predicted the considered instances, identifying both defective insulator disks and functional ones in the majority of the images.

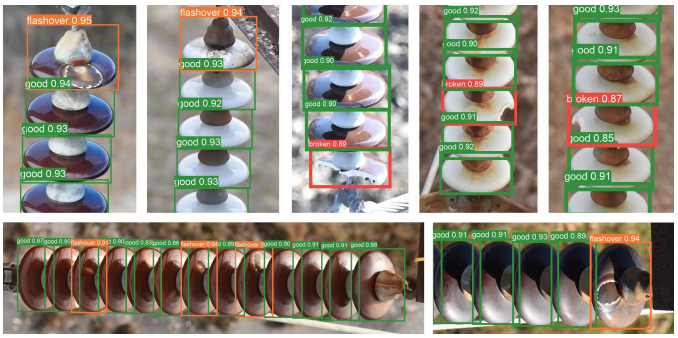


Fig. 5. Correct predictions on inference test using YOLOv8s



Fig. 6. Correct predictions on inference test using Faster R-CNN

By assessing some examples of mispredictions, it is possible to find patterns in the characteristics of the images and the objects wrongly detected or classified.

The 1st image in Figure 7 indicates that the YOLO model has difficulties identifying the actual texture of a broken insulator disk. The 2nd and 3rd images show that insulator disks contaminated with bird droppings alike are easily confused with flashover defects. In the 4th image, a defective insulator disk is correctly identified but with the wrong defect class. The presence of a power line in the foreground partially obstructing the object's view can impact the class prediction, as illustrated in the 5th image. In the 6th image is an object with multiple detections, having similar bounding boxes but different classification results. The 7th and 8th images show a flashover prediction due to unusual textures. Defocused objects in the 9th image are considered background, mainly because the training data consists of close planes with the focal point on the object's surface. The last image suggests that other elliptical shape objects can be misclassified as defective discs.

The first 3 images in Figure 8 suggest that the Faster model struggles to detect broken disks when these do not have a brighter, sharp edge visible. The second group of 2 images exemplifies that *good* insulators next to *broken* ones are often not detected. This trend aligns with the annotation practices adopted. Illustrated by the 6th and 7th images are objects with multiple detections, having similar bounding boxes but different classification results. If the main goal is identifying images with defects, this issue may not be a problem. Next, the misclassification of a *good* insulator as a *broken* one suggests that the power line in the foreground might have influenced the class prediction. The model is clearly not well-fitted yet for evening light conditions and prominent shadows due to a lack of training examples (e.g. 8th image). Lastly, translucent glass insulators cannot be correctly detected, due to the absence of glass insulators in the training dataset.

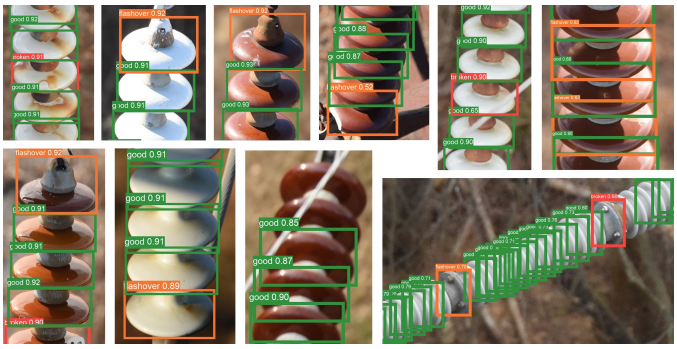


Fig. 7. Mispredictions on inference test using YOLOv8s

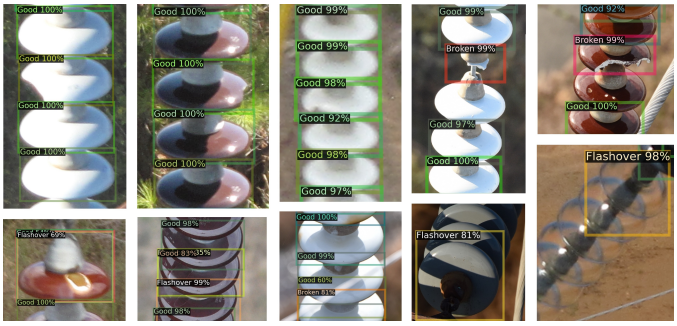


Fig. 8. Mispredictions on inference test using Faster R-CNN

Another non-annotated dataset for MV power lines, owned by EDP LabElec, was used for testing the model capabilities. Some images of this dataset were subjected to inference by both models, validating the potential of the approach conducted in this work in detecting and classifying disc insulator objects. Cropped images resulting from YOLO inference on the EDP LabElec dataset are presented in Figure 9.



Fig. 9. Example inference test for EDP LabElec data using YOLOv8s

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This inference test on a dataset with different characteristics from the one used for training revealed that the trained YOLO model achieved better generalization capabilities. The Faster model struggled to make valuable detections on images with blurred objects, primarily because the training data consists of close planes with the focal point on the object's surface. One future development is to perform enough expert annotations to create a large-scale training set that will enable to improve the model through a fine-tuning process.

VI. Conclusions and Future Work

This paper evaluated the potential of deep vision models to be used as a component for automated visual inspection of electrical grid assets, focusing on HV insulators. This performance analysis is conducted as a first step for the development of low complexity models targeting the integration in smart grid operations, aiming to improve the maintenance efficiency of electrical T&D systems. This inspection concept used for HV insulators can be expanded to other grid assets.

Defects in HV insulators were detected using SOTA deep learning models to process visible light images sourced from a publicly available dataset, that contains 1600 images with over 17000 annotations of insulator discs. Both models presented good performance results, for a fully automated approach, with a mean average precision (mAP@50), for the three considered classes (*good* insulator; *broken* disc; and *flashover* shell), of 87,9% and 87,2% on the YOLOv8s and Faster R-CNN X101-FPN, respectively.

For future research, in collaboration with EDP LabElec, we plan to proceed in there main directions: create a high-quality dataset, ensuring accurate annotations to improve model generalizability and robustness; improvements to the models, by maximising field knowledge and designing domain-specific architectures for the semantic hierarchic structure that characterises electrical grid assets; develop a low complexity model that can be used for edge computing and smart grid integration.

Numerous smart grid applications have embraced big-data and data-driven methodologies. However, the performance and complexity of the involved techniques must be thoroughly evaluated for their progressive integration towards real-time monitoring, control and maintenance management, in the context of digital twins and the smart grid. This is especially true for inspections using image sensors of various modalities, given the huge amount of data that can be involved in acquisition and edge processing.

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